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Monocular Vision for Robot Navigation

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***Abstract*— The structure and camera pose obtained using multiple view geometry based techniques cannot readily be used for robot localization and mapping. This is due to the fact that the structure and pose obtained relate to the actual environment and motion only up to a transform. In this paper, a method to localize the robot using monocular vision is presented. The assumptions are that the initial pose of the robot is known and that five or more landmarks (true, world points) can be identified. If two or more dissimilar views of at least five non coplanar feature points are initially available, subsequent robot locations with respect to the landmarks in view can be established. The exploration of the environment can then take place incorporating new feature points as the robot moves and successive images are acquired. The feature points which are no longer present in the field of view have to be handled along with the occluded ones. In the presented method, the recovered structure and the knowledge about the intrinsic parameters of the camera are used to obtain the metric structure. Depending on the number of images considered at a time, the structure recovery can be done using the epipolar constraints or using the factorization method. The coordinates of the known landmarks are used to calculate the true 3D world coordinates of the feature points. Current location of the robot is established with respect to these landmarks. The world coordinates of the subsequently observed feature points are obtained using the full camera calibration available following the robot localization. The proposed method avoids cumbersome stereo rig calibration. It naturally uses the new feature information available as the robot moves, for incremental localizations. The performance of the algorithm is verified with simulation and real results.**

***Index Terms*— multiple view geometry, metric reconstruction, world coordinates, epipolar geometry, factorization method, landmark based navigation**

I. Introduction

Vision is extensively being used for autonomous robot navigation as it offers a 'rich' sensory system. A large number of vision based mobile robot navigation schemes have been reported and interested readers are referred to the survey by DeSouza and Kak [1]. If a stereo-rig is used, triangulation can provide reasonably accurate range information for the environment mapping [2]. However, monocular vision (or in general multiple images using uncalibrated cameras) cannot provide actual 3D position data with respect to a reference (world) coordinate frame [3]. The ambiguity of structure available from multiple view geometry techniques is characterized by the unknown transform that maps the structure to the actual world coordinates of the feature points. Due to this limitation, it is not attractive to use a single camera on a vision guided robot. The advantages of using a single camera are the ability to efficiently make use of the already established extensive multiple view geometry techniques [3], [4]. As the robot moves, new images are acquired and new information becomes available. The incremental nature of information can be incorporated elegantly with existing multiple view geometry techniques. To make use of these rich techniques it is necessary to devise methods of relating the structure to the actual world coordinates.

Attempts of using a single camera for robot navigation have been reported for obstacle avoidance and self-localization. However, none of them make use of the possibility of obtaining the structure from multiple images and most rely on strong assumptions on the environment model. Problems addressed are locating the obstacles in the image and placing them on a map using real-world coordinates (calculating range data) [5]. The orientation of the camera with respect to the robot is assumed to be known and followed by a camera calibration process using a grid drawn on the ground and geometrical relations of subtended angles the range data vales are calculated [6]. Localization based on known geometrical objects has also been reported [7]. A different approach is using optical flow techniques for navigation and obstacle avoidance [8]. There are extended Kalman filter based stochastic methods proposed for navigation using vision to a certain degree, to provide location information [2], [9]. The technique is to model the robot pose as the 'state' [10] and use recursion for updating the state having modeled the robot motion as the state transition. The use of vision as the primary driving force is minimal.

The nature of the algorithm proposed in this paper makes it possible for many existing multiple view geometry techniques to be effectively used for robot localization and navigation. No assumptions are made on the robot motion. In the proposed method the robot localization at every robot position is carried out in three steps. At the first step features are tracked over a specified number of most recent images. The second step is performing the 3-D reconstruction and

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upgrading the reconstructed features to the true coordinates using the knowledge of the current landmarks. Finally, the transform obtained in the second step and the knowledge on the initial position of the robot are used to obtain the true robot position (and orientation). The algorithm proceeds by declaring the features upgraded to the true world coordinates as new landmarks and the estimated robot position as the new initial position. In the subsequent discussion, the terms true coordinates, world coordinates, ground control points and landmarks are synonymously used to identify points with known locations in Euclidean world frame and the robot pose [11] and camera pose are used synonymously.

II. 3-D Reconstruction

3-D reconstruction or structure from motion takes a set of images of a scene as the input and produces the 3-D structure and the camera pose up to a projective transform in general. Detailed descriptions can be found in literature ([12], [3], [13], [14], [15], [16]). A camera matrix in general is is composed as

$$\mathtt{P} = \mathtt{K}[\mathtt{R}|\mathbf{t}] \tag{1}$$

where $\mathtt{K}$ is the camera calibration matrix, $\mathtt{R}$ is a rotation matrix and $\mathbf{t}$ is a translation vector. Together, $\mathtt{R}$ and $\mathbf{t}$ define the external camera calibration or the robot pose [11], [17].

An image is obtained through a mapping from the 3-dimensional object space to the 2-dimensional image space, using the following form

$$\begin{bmatrix} u_{ij} \\ v_{ij} \\ 1 \end{bmatrix} \approx \mathtt{P}_i \mathbf{X}_j \tag{2}$$

Here, the approximate equality represents the associated scaling effects [3]. The measurement matrix $\mathtt{W}$ of m feature points across n images is

$$\mathtt{W} = \begin{bmatrix} \lambda_{11}\begin{bmatrix} u_{11} \\ v_{11} \\ 1 \end{bmatrix} & \lambda_{12}\begin{bmatrix} u_{12} \\ v_{12} \\ 1 \end{bmatrix} & \dots & \lambda_{1m}\begin{bmatrix} u_{1m} \\ v_{1m} \\ 1 \end{bmatrix} \\ \lambda_{21}\begin{bmatrix} u_{21} \\ v_{21} \\ 1 \end{bmatrix} & \lambda_{22}\begin{bmatrix} u_{22} \\ v_{22} \\ 1 \end{bmatrix} & \dots & \lambda_{1m}\begin{bmatrix} u_{1m} \\ v_{1m} \\ 1 \end{bmatrix} \\ \vdots & \vdots & & \vdots \\ \lambda_{n1}\begin{bmatrix} u_{n1} \\ v_{n1} \\ 1 \end{bmatrix} & \lambda_{n2}\begin{bmatrix} u_{n2} \\ v_{n2} \\ 1 \end{bmatrix} & \dots & \lambda_{nm}\begin{bmatrix} u_{nm} \\ v_{nm} \\ 1 \end{bmatrix} \end{bmatrix} \tag{3}$$

Here, $[u_{ij}, v_{ij}, 1]^T$ represents the homogeneous image coordinates of j-th feature point in the i-th image in terms of pixels. $i = 1, \cdots, n$ and $j = 1, \cdots, m$. This is the measurement matrix for the projective case and it is scaled by the projective depths λ_{ij} leading the problem to become non-linear.One approach to solve the problem is to recover the projective depths using fundamental matrices and epipoles [18], [14]. The other approach is to recover depths in an iterative manner [14], [19], [20]. In such iterative schemes, camera calibration and Euclidian structure recovery can simultaneously be done by enforcing Euclidian constraints on the camera calibration parameters [21], [13].

A. Structure and Motion Recovery

The measurement matrix $\mathtt{W}$ can be factorized into two matrices spanning the column space (motion) and the row space (shape) using the singular value decomposition (SVD) [22], [23] as

$$\mathtt{W} = \mathtt{U}\mathtt{S}\mathtt{V}^T \tag{4}$$

where $\mathtt{S}$ is a diagonal matrix formed of the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$, $r = \min(m, n)$. Tomasi and Kanade [24] pointed out that for the orthographic case, $\mathtt{W}$ can at most be of rank 3 and for the projective case rank 4. Now motion and shape can be constructed using the diagonal matrix composed of the first 4 singular values $\hat{\mathtt{S}}$ as,

$$\begin{aligned} \hat{\mathtt{P}} &= \mathtt{U} \\ \hat{\mathbf{X}} &= \hat{\mathtt{S}}\mathtt{V}^T \end{aligned} \tag{5}$$

Note that, this decomposition is not unique as any nonsingular matrix of appropriate size would satisfy $\mathtt{W} = \hat{\mathtt{P}}\mathtt{H}\mathtt{H}^{-1}\hat{\mathbf{X}}$. The process of resolving this ambiguity is called the metric reconstruction discussed in the following Section II-B.

B. Metric Reconstruction and Using Landmarks

The term *metric reconstruction* implies that the structure is defined up to a similarity. A similarity is a transformation of the form

$$\mathtt{H}_s = \begin{bmatrix} s\mathtt{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \tag{6}$$

where $\mathtt{R}$ is a rotation matrix and s is a scaling factor [3]. There are several approaches to obtain the metric reconstruction from the projective structure. The process called stratification first starts with a projective solution and progressively refines to affine and then finally to metric solution [25]. The alternative called the direct method [3] is to use the ground truth and jump directly form projective reconstruction to true reconstruction. The ground truth used is a set of ground control points (known landmarks). These are points with known 3D locations in the world frame. The final reconstruction obtained relates to the world using which the landmarks are specified. Consequently, at this point the structure can readily be used for robot localization.

If there are n such ground control points, $\mathbf{X}_t$, $i = 1, \cdots, n$, the 3D location $\mathbf{X}_i$ of these points in the projective reconstruction can be related using the projectivity $\mathtt{H}$ as [3]

$$\mathbf{X}_t = \mathtt{H}\mathbf{X}_i \quad i = 1, \cdots, n \tag{7}$$

Each point correspondence results in 3 linear equations on the elements of $\mathtt{H}$. As $\mathtt{H}$ has 15 degrees of freedom a linear

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solution can be obtained if $n \geq 5$ provided that no four of the landmarks are coplanar. The camera matrix (and the robot location and orientation up to a scale) can be obtained as

$$\mathtt{P}_{ts} = \mathtt{P}\mathtt{H}^{-1}. \tag{8}$$

1) Projectivity H: The projectivity $\mathtt{H}$ is found by observing that $\mathtt{H}\mathbf{X}_i$ is (approximately) parallel to $\mathbf{X}_{ti}$. Consequently, any orthogonal vector to $\mathbf{X}_{ti}$ denoted by $\mathbf{X}_{ti}^{\perp}$ should be orthogonal to $\mathtt{H}\mathbf{X}_i$ as well, i.e.,

$$\mathbf{X}_{ti}^{\perp}\mathtt{H}\mathbf{X}_i \simeq 0 \tag{9}$$

where

$$\mathbf{X}_{ti}^{\perp} = \frac{\mathbf{I} - \mathbf{X}_{ti}\mathbf{X}_{ti}^{T}}{\mathbf{X}_{ti}^{T}\mathbf{X}_{ti}} \tag{10}$$

Each equation of the form of the Equation 9 corresponding to a feature point gives rise to 4 linear equations of which 3 are independent. Equation 9 can be rewritten as a matrix by stacking vectors generated by the Kronecker product [12] between $\mathbf{X}_{ti}^{\perp}$ and $\mathbf{X}_i$. Stacking 5 or more such matrices, each corresponding to a feature point i, and and solving the resulting system computes the elements of $\mathtt{H}$. Although 5 points are sufficient in theory, it is advantageous to use more than five points and to obtaing the least squares solution using the SVD [23].

III. Method

The landmark based localization is performed as each new image is acquired (i.e. at each subsequent robot position). The method consists of feature tracking, reconstruction, upgrading to true coordinates, calculating the robot position and propagating the landmarks and the new robot position. As the first step features are tracked over a specified number of most recent images. In the current implementation only 2 or 3 images are used for simplicity. The second step is performing the 3-D reconstruction and upgrading the reconstructed features to the true coordinates using the knowledge of the current landmarks. Once the projectivity $\mathtt{H}$ (Section II-B.1) is known the feature points of which the 3-D world coordinates are not available can also be transformed onto the world frame. Finally, the projectivity obtained in the second step and the knowledge on the initial position of the robot are used to obtain the true robot position. The algorithm proceeds by declaring the features upgraded to the true world coordinates as new landmarks and the estimated robot position as the new initial position.

A. Reconstruction and Upgrading to True Coordinates

The 3-D reconstruction is done using the factorization method outlined in the Section II-A using n images at each robot position. n is typically a small number. It is important to scale the measurement matrix so that each homogeneous image coordinate is of unity norm for the SVD to be well conditioned [14], [3]. The assumption that at least five feature points that appeared in the previous image are still present in the current image enables the re-computation of the robot pose. This constraint requires sufficient number of feature points to be available and the robot motion to be small between two images. The computation of the projectivity $\mathtt{H}$ is done as the mapping from the reconstructed features to their true counterparts as described in the Section II-B.1.

Now the true coordinates of all the feature points visible in the first set of images can be determined using

$$\mathbf{X}_t = \mathtt{H}\mathbf{X}_i \tag{11}$$

The camera matrices are computed up to a scale as

$$\mathtt{P}_{ts} = \mathtt{P}\mathtt{H}^{-1} \tag{12}$$

The robot pose is found by multiplying the camera matrix by the inverse of the camera calibration matrix.

$$[\mathtt{R}_i|\mathbf{t}_i] = \mathtt{K}^{-1}\mathtt{H}_{ti} \tag{13}$$

Camera calibration (obtaining $\mathtt{K}$) was done using the four step method proposed by Heikkila *et al.* [26]. The Equation 12 can only map the recovered camera poses to the true ones only up to a scale. The assumption that the first camera position is known enables this scale factor to be recovered. A simple method of obtaining the scale is using the element ratios as follows.

$$\mathbf{s} = \mathbf{t_t}/\mathbf{t_i} \tag{14}$$

The scale factor $\mathbf{s}$ is used to calculate the true robot position for the subsequent camera matrices.

$$\mathbf{t}_{t,i+1} = \mathbf{s}\mathbf{t}_{i+1} \tag{15}$$

where $\mathbf{t}_{t,i+1}$ is the (desired) next true robot position. If the number of images used in reconstruction is n, $n-1$ new robot positions can be recovered.

B. Algorithm

1) Find matching feature point in the set of n most recent images
2) Using the algorithms given by Triggs [14] obtain the factorization based reconstruction (structure and motion)
3) Isolate the reconstructed features corresponding to the known true coordinates (landmarks) and obtain the projectivity (Section II-B.1)
4) Find the true coordinates of all the feature points using the projectivity (Equation 11). These true coordinates are deemed to be landmarks in subsequent operations
5) Find the true location and orientation of the robot using the projectivity and the scale factor (Equations 13 and 14)
6) Obtain the next image, $(n+1)$, update the set of landmarks, update the initial robot position $\mathbf{t}_t$ and repeat from step 1

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IV. Simulations and Results

A. Simulation Deatails

Simulations were carried out using synthetic scenes. A scene consisted of points with known true coordinates. This knowledge is required just for the purpose of making synthetic images using a predefined camera model. The camera model used was characterized by the camera calibration matrix [3]

$$\mathtt{K} = \begin{bmatrix} \alpha_x & s & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1000 & 0 & 256 \\ & 1000 & 256 \\ & & 1 \end{bmatrix} \tag{16}$$

assuming an image of size 512×512 pixels and focal lengths of $\alpha_x = \alpha_y = 1000$ pixels. The images were made using

$$\begin{bmatrix} u_{ij} \\ v_{ij} \\ Z \end{bmatrix} = \mathtt{P}_i \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix} \tag{17}$$

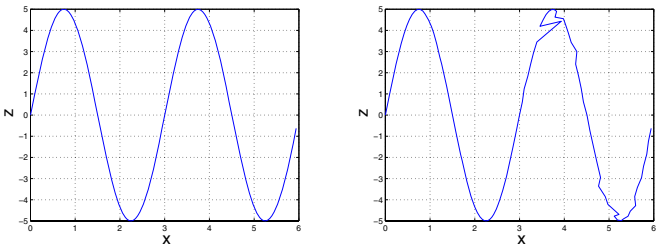
$$\mathbf{x}_{ij} = \mathtt{P}_i \mathbf{X}_{tj}$$

where $\mathtt{P_j}$ was the jth camera and $\mathbf{X}_{tj}$ is the jth true feature point. The measurement matrix was constructed in the form of Equation 3. Gaussian random noise was added to each u_{ij} and v_{ij}. The algorithm given in Section III-B was employed at the acquisition of each successive image. The factorization based reconstruction was done using the algorithm in Triggs [14] and the result at this point was similar to Equation 5. It is important to scale the measurement matrix $\mathtt{W}$ for numerical stability of the algorithm [3], [14]. Assuming that an $n = 8$ number of true feature points are known, the homography $\mathtt{WH}$ was calculated. Once the homography was available true robot pose was computed using Equations 13 and 14.

B. Results with Synthetic Images

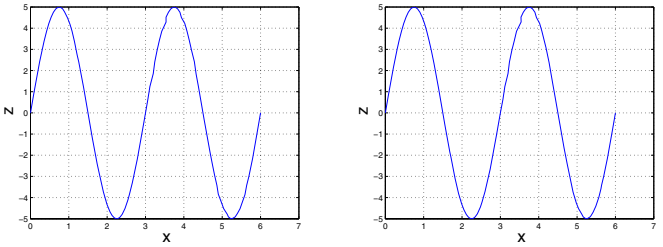
Simulations were carried out using two, there and four views at each step, with a sinusoidal motion of the robot. As the robot traversed sinusoidally the camera orientation was also changes in a sinusoidal fashion. The variance of Gaussian random noise added was in the orders of 10^{-6} to $10^{-4.5}$. The Localization simulation using only two views is shown in Figure 1. The performance deteriorates drastically in the presence of pixel noise. However, when three images are used for reconstruction (Figure 2 and 3), the noise performance is superior. A better picture of noise performance can be obtained if more than one manifestation of noise process are used (ensemble averaged performance). Figure 4 shows the final (after 100 steps of robot motion) reconstructed version of a test scene of a road side.

Even though five landmarks are sufficient, in theory, for the algorithm, the results were inaccurate. The performance was satisfactory for six and higher number of landmarks. Figure 5 shows the location error between the actual robot location and the estimated location. These curves are for a single realization of the noise and hence the curves may vary with the noise realization.



(a) Noise $\sigma = 10^{-6}$ (b) Noise $\sigma = 10^{-5}$

Fig. 1. Localization performance with two views. The experiments were done using 100 motion steps in the form a sinusoid. Without the presence of image noise, the results are accurate. Noise drastically degrades the performance.



(a) Noise $\sigma = 10^{-6}$ (b) Noise $\sigma = 10^{-5}$

Fig. 2. Localization performance with three views. The experiments were the same as for Figure 1. Without the presence of image noise, the results are accurate. Noise performance is better than that of the two view case.

C. Results with Real Images

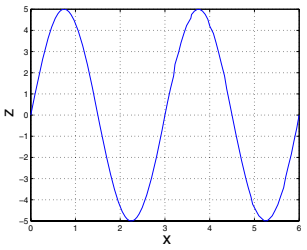
The number of images used for reconstruction at each robot position was $n = 3$. A sample image from the sequence is shown in Figure 6. The room was approximately 800 cm $\times$ 600 cm $\times$ 300 cm. The reconstruction is shown in Figure 7. While the reconstruction was reasonably accurate, the mean deviation from the true landmarks was 7.8604 cm.

The localization error was 22.8349 cm, 30.4051 cm 31.9157 cm and 39.5402 cm for a sample of four successive camera positions, within the same room mentioned above. The localization error error was due to the numerical ill-condition of the reconstruction and projectively calculation algorithms. In addition the fact that there is no error correction used in the scale factor calculation contributes to the error.

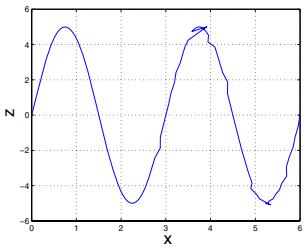
V. Conclusion

A method of using monocular vision for robot localization wad demonstrated. The algorithm effectively makes use of the existing rich multiple view geometry techniques. If five or more landmarks of known true world coordinates are visible in two or more images, and if the initial camera

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(a) Noise $\sigma = 10^{-5}$



(b) Noise $\sigma = 10^{-4.5}$

Fig. 3. Localization performance with four views. The experiments were the same as for Figure 1. The noise performance is better

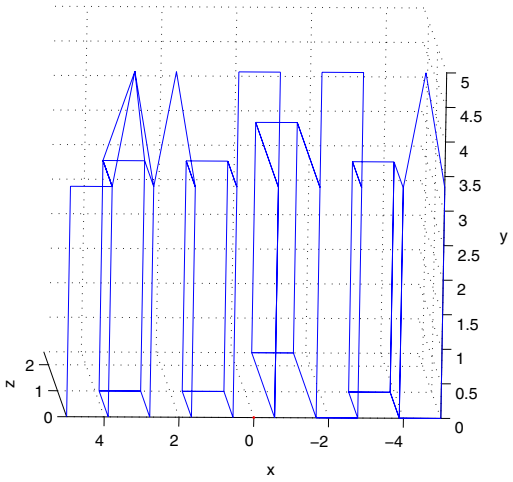
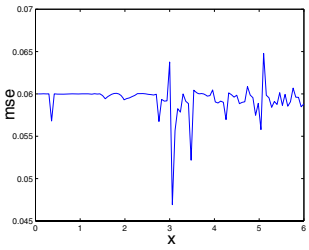


Fig. 4. Final reconstructed structure for a test scene, a roadside. Noise $\sigma = 10^{-6}$

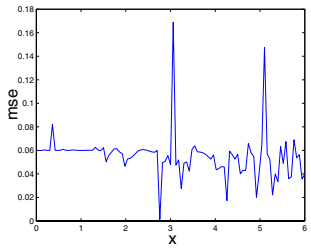
pose is known, monocular localization is possible. However, six or more landmarks are desirable for accurate localization in the presence of noise. Even though the the performance drastically degrades in the presence of noise, as the number of images used for reconstruction increases, the noise performance improves.

A. Limitations

It is assumed that at least five feature points (landmarks) that appeared in the previous image are still present in the current image to have a sufficient number of landmarks visible in every image. The assumption is more stringent in the multiple view case when all the images considered should include all the landmarks. A further assumption that the initial robot pose is known is required to scale the reconstructed robot poses up to the true poses. However, in a typical indoor environment, meeting these assumptions is not difficult. At each stage the localization relies on the landmarks obtained from the previous step. Therefor, the error accumulation is significant. This problem can be solved if additional landmarks become available as the robot moves



(a) Noise $\sigma = 10^{-6}$



(b) Noise $\sigma = 10^{-5}$

Fig. 5. Mean squares error between estimated location and the actual location of robot for the 100 steps

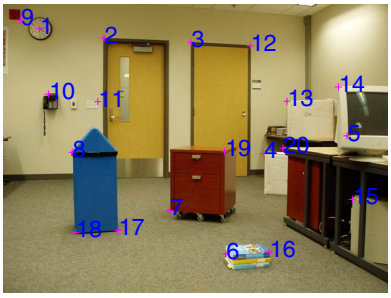


Fig. 6. Features used in reconstruction. Features 1-8 are assumed to be landmarks.

or if a prediction-correction method is employed to reduce uncertainty.

B. Future Work

Given the current position and the robot motion, it is possible to predict the next pose of the robot. Therefore, the camera can be considered to be fully calibrated with uncertainties. This work does not make use of this information to come up with a stochastic algorithm to localize the robot. A further simplification assumed in the implementation is that the intrinsic camera parameters (the camera calibration matrix) is known. Autocalibration techniques can be incorporated with the proposed method without difficulty. The algorithm has to be tested with real images and handling occlusions will be a challenge.

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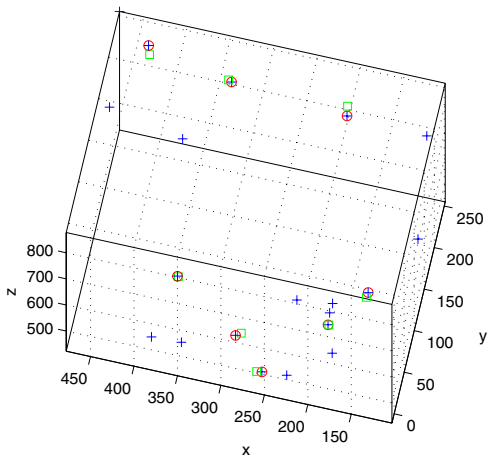


Fig. 7. Reconstruction after upgrading to true coordinates using landmarks. □ : landmarks (ground truth), ○ : reconstructed landmarks + : all reconstructed features

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